

AIOps Engineer

Complete Training Syllabus · Updated 2026

🕒 Duration: 12 Weeks

📜 Certificate: GenZNext Certified AIOps Engineer

🔑 Level: Intermediate–Advanced

01

MODULE 01

Foundations of AIOps

- AIOps Philosophy: From Reactive to Predictive Ops
- AIOps vs. Traditional Monitoring vs. MLOps
- Core Pillars: Data, Algorithms & Automation
- IT Operations Lifecycle & Where AI Fits In
- Key Metrics: MTTR, MTTD, Alert Noise Ratio
- Building an AIOps Maturity Roadmap

02

MODULE 02

Observability & Telemetry Engineering

- Logs, Metrics, Traces & Events — the Four Signals
- Multi-Source Telemetry Ingestion Pipelines
- Prometheus, Grafana & the Elastic Stack
- Data Quality, SLOs & Lineage for Telemetry
- Streaming Pipelines with Kafka & Airflow
- Building a Unified Observability Data Lakehouse

03

MODULE 03

Machine Learning for Anomaly Detection

- Statistical vs. ML-Based Anomaly Detection
- Dynamic Thresholding (Replacing Static Alerts)
- Time-Series Forecasting for Capacity & Failures
- Training & Registering Models (MLflow, TensorFlow)
- Evaluating Models Against Labelled Incident Data
- Reducing False Positives & Alert Fatigue

04

MODULE 04

Event Correlation & Noise Reduction

- Event Correlation Engines (Moogsoft-style Patterns)
- Topology-Aware & Causal Correlation
- Deduplication & Alert Grouping Strategies
- Building Correlation Rules vs. Learned Correlation
- Routing Correlated Incidents to PagerDuty/On-Call
- Measuring Signal-to-Noise Improvement

05

MODULE 05

Automated Root Cause Analysis

- RCA Fundamentals: Symptoms vs. Root Cause
- Graph-Based & Dependency-Aware RCA
- Log Pattern Mining for Root Cause Signals
- Causal Inference Techniques for Incidents
- RCA Copilots: Assisting the On-Call Engineer
- Postmortems & Feeding RCA Back into Models

Self-Healing & Automated Remediation

- Designing Self-Healing Runbooks
- Auto-Remediation Triggers & Guardrails
- Kubernetes-Native Healing (Auto-Scaling, Restarts)
- Closed-Loop Automation: Detect → Decide → Act
- Safety Nets: Approval Gates for High-Risk Actions
- Chaos Engineering to Validate Self-Healing

GenAI & LLM-Powered Operations

- LLM-Powered Incident Diagnostics & Summarization
- Natural-Language Querying of Telemetry Data
- AI Agents for On-Call Assistance & Ticket Triage
- Retrieval-Augmented Runbook Generation
- Prompting Patterns for Ops Copilots
- Guardrails & Cost Control for LLM-Ops Workloads

AIOps Platform Engineering & MLOps

- Deploying AIOps Models via CI/CD Pipelines
- Model Serving, Monitoring & Drift Detection
- Security, RBAC & Governance for AIOps Platforms
- Multi-Cloud AIOps: AWS, Azure & GCP Patterns
- Cost Optimization for AI-Driven Ops (FinOps Tie-In)
- Capstone: End-to-End AIOps System with Postmortem Drill

What You'll Walk Away With

A practitioner-grade path from telemetry and anomaly detection to self-healing systems and LLM-powered ops copilots
— built to make you production-ready, not just certificate-ready.

[Live Mentorship](#)[Real Telemetry Labs](#)[Capstone Project](#)[LMS Access for Life](#)